Incident Ticket #4: Incorrect DHCP Subnet & Gateway Assignments

Status: Resolved

Skill Evaluation: Strong diagnostic process; successfully separated an access-port VLAN misconfiguration from a DHCP scope default-gateway error.

Ticket Overview

- **Ticket ID:** TCK-004
- **Affected Hosts:** Reception PC 1 & Admin PC
- **Affected Services:**
- DHCP
- VLAN 10 Reception_Data
- VLAN 20 Admin_Data
- **Symptom:** Reception PC 1 received an IP address and default gateway on the wrong subnet. Admin PC received a valid IP address on its subnet, but was handed an incorrect default gateway.
- **Expected Behavior:**
    - Reception PC 1 $\rightarrow$ Subnet 192.168.10.0/24, Gateway 192.168.10.1 (VLAN 10)
    - Admin PC $\rightarrow$ Subnet 192.168.20.0/24, Gateway 192.168.20.1 (VLAN 20)
- **Actual Behavior:**
    - Reception PC 1 $\rightarrow$ Received IP 192.168.20.12 and Gateway 192.168.25.1
    - Admin PC $\rightarrow$ Received IP 192.168.20.11 and Gateway 192.168.25.1

Diagnostic Matrix & Root Causes

Host / Target	Affected Layer	Discovered Condition	Root Cause	Corrective Action

Reception PC 1	Layer 2 Access Port	Switch A FastEthernet0/4 set to switchport access vlan 20	Port was in VLAN 20 instead of VLAN 10, placing DHCP requests in the wrong broadcast domain	Reassigned Fa0/4 to access VLAN 10
Admin PC	Layer 3 DHCP Server	Router pool ADMIN_POOL configured with default-router 192.168.25.1	Router DHCP pool handed out Doctors VLAN gateway (192.168.25.1) to VLAN 20 clients	Reconfigured ADMIN_POOL default router to 192.168.20.1

Note: Reception PC 1 originally received a Gateway of 192.168.25.1 because its port was misconfigured to VLAN 20, inheriting the flawed DHCP gateway setting from the Admin pool.

Step-by-Step Resolution

```
[Issue 1: Switch A Fa0/4]                 [Issue 2: ROAS Router]
  Assigned to VLAN 20                        ADMIN_POOL Default Router: 192.168.25.1
        |                                |
        v                                 v
  Fix: Reassign Fa0/4 to VLAN 10              Fix: Set default-router to 192.168.20.1
        |                                |
        +----------------------+------------------------+
                               |
                               v
                   [ipconfig /release & /renew]
```

Step 1: Fix Switch A Access Port (Reception PC 1)

Switch-A> enable
Switch-A# configure terminal
Switch-A(config)# interface FastEthernet0/4
Switch-A(config-if)# switchport mode access

Switch-A(config-if)# switchport access vlan 10
Switch-A(config-if)# switchport voice vlan 30
Switch-A(config-if)# no shutdown
Switch-A(config-if)# end

Step 2: Renew Reception PC 1 IP Lease

C:\> ipconfig /release
C:\> ipconfig /renew

IPv4 Address. . . . . : 192.168.10.12
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 192.168.10.1

Step 3: Correct Router DHCP Pool Configuration (Admin PC)

ROAS# configure terminal
ROAS(config)# ip dhcp pool ADMIN_POOL
ROAS(dhcp-config)# network 192.168.20.0 255.255.255.0
ROAS(dhcp-config)# no default-router 192.168.25.1
ROAS(dhcp-config)# default-router 192.168.20.1
ROAS(dhcp-config)# end
ROAS# copy running-config startup-config

Step 4: Renew Admin PC IP Lease

C:\> ipconfig /release
C:\> ipconfig /renew

IPv4 Address. . . . . : 192.168.20.11
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 192.168.20.1

Verification Proof

Reception PC 1:
  IPv4 Address....................: 192.168.10.12
  Subnet Mask.....................: 255.255.255.0
  Default Gateway.................: 192.168.10.1

- Reception PC 1 → 192.168.10.1 (Gateway): Success

Pinging 192.168.10.1 with 32 bytes of data:
    Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

- Reception PC 1 → Admin PC 192.168.20.11: Success

Pinging 192.168.20.11 with 32 bytes of data:
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time=1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

Admin PC:
IPv4 Address....................: 192.168.20.11
Subnet Mask.....................: 255.255.255.0
Default Gateway.................: 192.168.20.1

- Admin PC → 192.168.20.1 (Gateway): Success

Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

- Admin PC → Reception PC 1 192.168.10.12: Success

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=127
Reply from 192.168.10.12: bytes=32 time<1ms TTL=127
Reply from 192.168.10.12: bytes=32 time=2ms TTL=127
Reply from 192.168.10.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.12:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 2ms, Average = 0ms

Key Technical Takeaways

- When a host receives an unexpected IP from an entirely different subnet, verify its Layer 2 switchport access VLAN configuration first before altering Layer 3 DHCP settings.
- When a host receives a valid IP in the correct subnet but the wrong default gateway, inspect the DHCP pool configuration with show running-config or verify scope behavior with show ip dhcp pool to look for default-router parameter typos.